Resources for Interlingual machine translation system

- Dictionaries (For analysis and generation)
- A conceptual knowledge based lexicon
- A set of projection rules
- Grammars for the analysis and generation of the languages involved

Characteristics of Interlinguas

- Unambiguous
  - should be more explicit.
- Represent the full content of the input text
  - Morphological
  - Syntactic
  - Semantic
  - Pragmatic char c
- should be Universal
  - Abstract meaning of any text.
- should represent the content of the input alone
- should be independent of both the source language text and the target language text.

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Interlingua:

Interlingua machine translation is a classical approach.

Two steps of Interlingua:

1. Source language text is transformed into an interlingua (Here, interlingua is an abstract language-independent representation)
2. Target language is generated from the Interlingua.

[Alternative machine translation approach to the direct approach and the transfer approach]

Advantages of Interlingual approach:

- Requires fewer components in order to relate each source language to each target language.
- Takes fewer components to add a new language.
- Supports paraphrases of the input in the original language
- Allows both the analysers and generatoos to be written by monolingual system developers.
- Handles languages that are very different from each other

Disadvantages of Interlingual approach:

- Definition of an interlingua is difficult.

Objective of Interlingua approach:

Analysis of text of a source language with respect to

i) Morphology
ii) Syntax
iii) Semantic

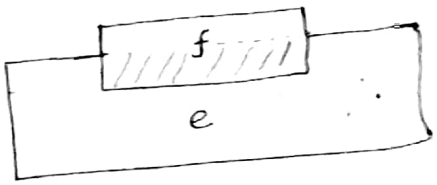
} Meaning

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that the best translation is the sentence that is both faithful to the original sentence and fluent in the target language

Ans:

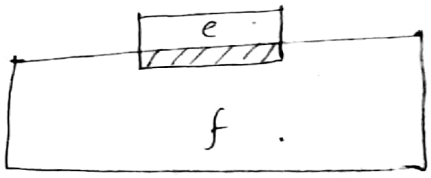
$$P(f|e) = \frac{\text{Total } f \text{ in } e}{\text{Total } e}$$



$$= \frac{P(f \wedge e)}{P(e)}$$

$$\Rightarrow P(f \wedge e) = P(f|e)\, P(e) \quad \cdots \quad eq^{(1)}$$

$$P(e|f) = \frac{\text{Total } e \text{ in } f}{\text{Total } f}$$



$$= \frac{P(f \wedge e)}{P(f)}$$

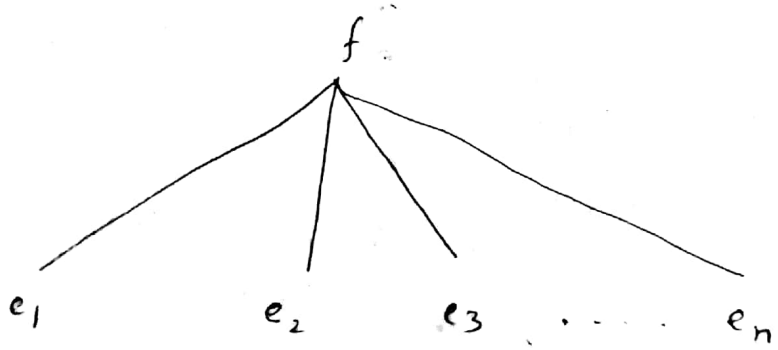
$$\Rightarrow P(f \wedge e) = P(e|f)\, P(f) \quad \cdots \quad eq^{(2)}$$

From $eq^{(1)}$ and $eq^{(2)} \Rightarrow$

$$P(e|f)\, P(f) = P(f|e)\, P(e)$$

$$\Rightarrow P(e|f) = \frac{P(f|e)\, P(e)}{P(f)}$$

If there are n sentences in target language for source language f



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$$P(e_1|f) = \frac{P(f|e_1)\ P(e_1)}{P(f)}$$

$$P(e_2|f) = \frac{P(f|e_2)\ P(e_2)}{P(f)}$$

$$P(e_3|f) = \frac{P(f|e_3)\ P(e_3)}{P(f)}$$

$$P(e_n|f) = \frac{P(f|e_n)\ P(e_n)}{P(f)}$$

$$P(e|f) = \max\left\{ P(e_1|f),\ P(e_2|f),\ P(e_3|f),\ \ldots\ldots,\ P(e_n|f) \right\}$$

$$= \max\left\{ \frac{P(f|e_1)\,P(e_1)}{P(f)},\ \frac{P(f|e_2)\,P(e_2)}{P(f)},\ \ldots\ldots,\ \frac{P(f|e_n)\,P(e_n)}{P(f)} \right\}$$

$$\cong \max\left\{ P(f|e_1)\,P(e_1),\ P(f|e_2)\,P(e_2),\ \ldots\ldots,\ P(f|e_n)\,P(e_n) \right\}$$

$$= \max\left\{ P(f|e_i)\ P(e_i) \right\}$$

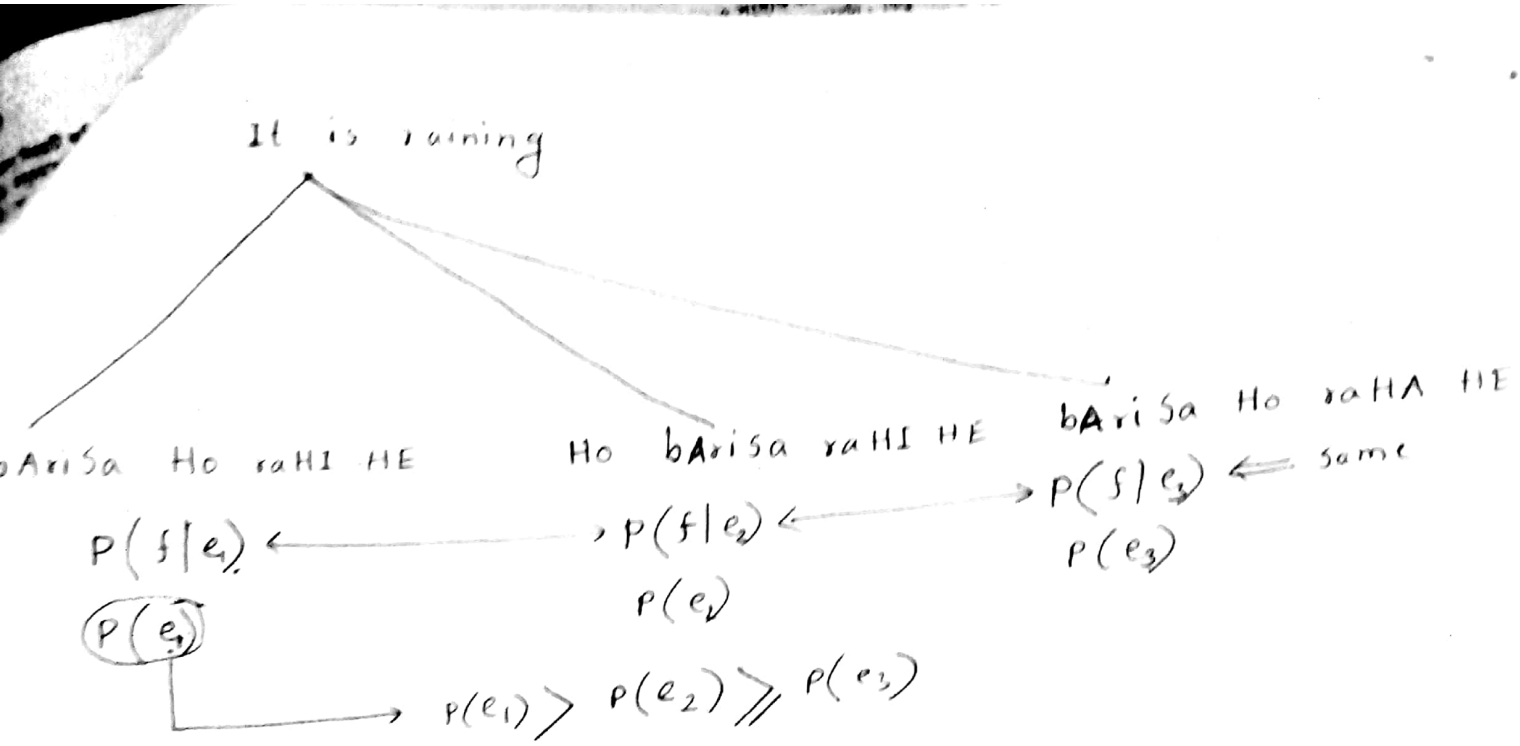
where $i = 1 \ldots n$

$$\hat{e} \in \{e_1, e_2, \ldots e_n\}$$

$$\hat{e} = \arg\max_{e_i} \left\{ P(f|e_i)\ P(e_i) \right\}$$

- $P(f|e_i)$ → Faithful sentence for source text
- $P(e_i)$ → fluent sentence in target language

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Since $P(f|e_1)\,P(e_1)$ maximum

It is raining $\Longrightarrow$ bArisa Ho raHI HE

Noisy channel Model for Machine Translation :

Let source sentence f to be translated was initially conceived in language E as some target sentences e.

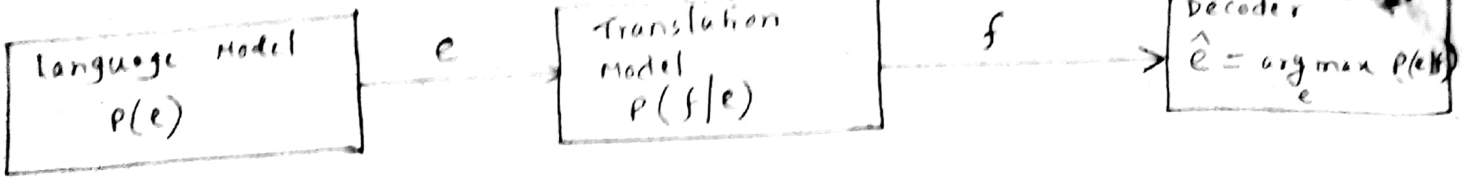
During communication e was corrupted by the channel to f

Each sentence in E is a translation of f with some probability

The sentence we choose as the translation ($\hat{e}$) is the one that has the highest probability

$$\hat{e} = \underset{e}{\arg\max}\ P(e|f)$$

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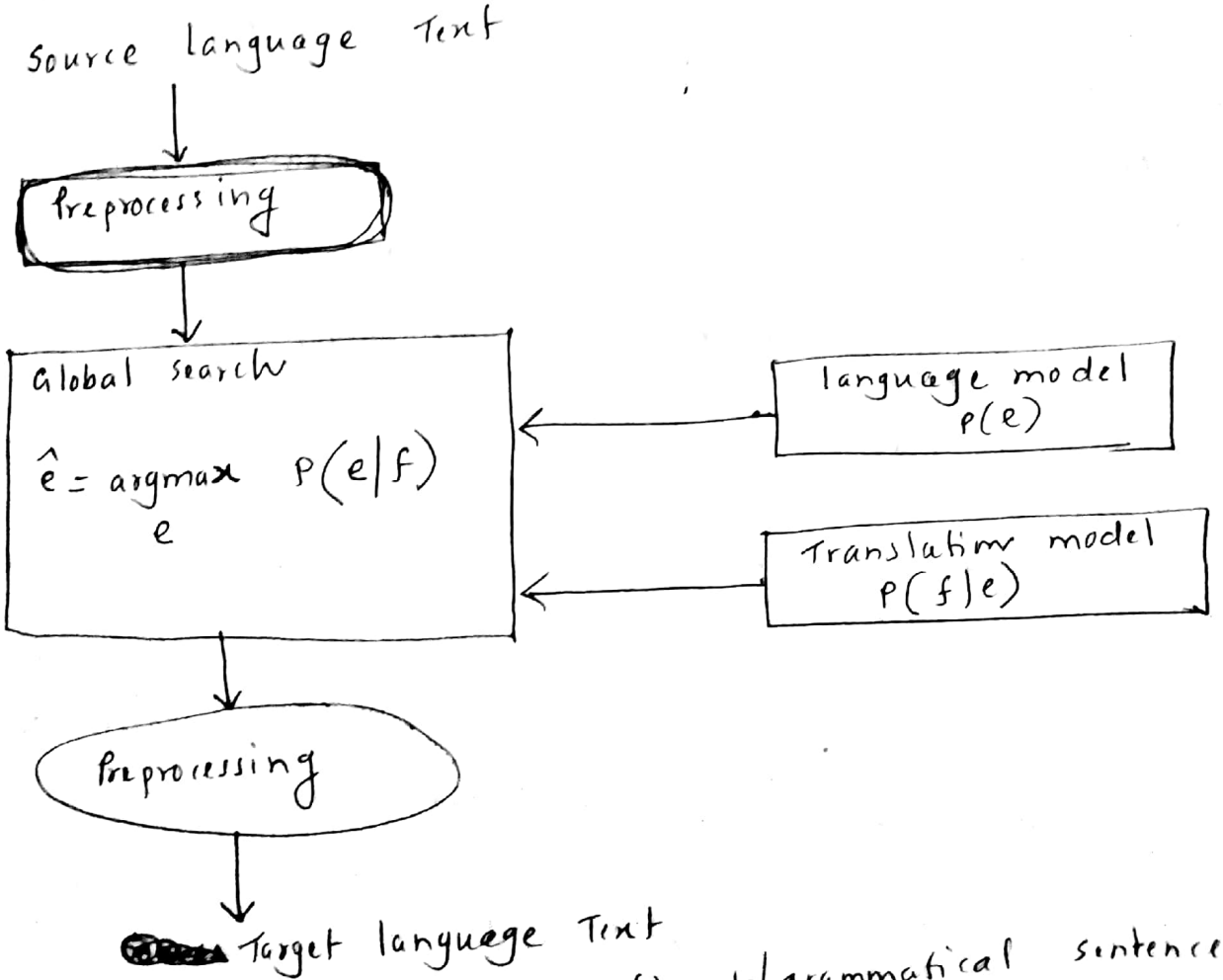
P(e): language model

The kind of sentences that are likely in the language E.

P(f|e): Translation model

The way sentences in E get converted to sentences in F. This is called translation model.

System diagram of Machine Translation:



P(e): Assigns higher probability to fluent/grammatical sentences
: Estimated using monolingual corpora

P(f|e): Assign higher probability to sentences that have corresponding meaning
: Estimated using bilingual corpora

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